MAIS PAC

IPhO 02.02: Statics: Forces, Torques and Equilibrium · Worked Solutions

Statics: Forces, Torques and Equilibrium: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IPhO questions or marking schemes. Parameters are supplied locally; an exam’s constants take precedence.

Solution index

1 2 3 4 5 6 7 8 9 10

Solution 1 — A balanced rule. Return →

(a) The rule's own weight acts at the 50 {cm} mark, i.e. at the pivot, so it contributes no torque. Torque balance: 200(50-20) = 150(50-x) 6000 = 150(50-x) 50-x = 40, so x = 10 {cm}.

(b) Vertical balance: N = (200 + 120 + 150)g = 470 × 9.81 = 4.61 {kN}.

(c) Moving the pivot to 30 {cm} puts the rule's weight 20 {cm} from the pivot, adding a torque that must be balanced; also the 200 {g} mass is now at the pivot and contributes nothing. The balance point changes and the reaction is unchanged at 4.61 {kN} because vertical forces have not changed. **Check:** distances from the pivot, not from the left end, are what torque balance uses.

Solution 2 — The ladder problem. Return →

(a) Let R_w be the wall reaction and N_f the floor reaction. Vertically N_f = mg. Torques about the foot: the wall reaction R_w acts at height Lsin θ with lever arm Lsin θ; the weight acts at the midpoint with horizontal lever arm (L)/(2)cos θ:

$$R_w \, L\sin\theta = mg(L)/(2)\cos\theta \quad R_w = (1)/(2)mg\theta. \tag{1}$$

(b) Horizontally, f = R_w = (1)/(2) mgθ.

(c) Slip occurs when f > μ N_f = μ mg, i.e. (1)/(2) mgθ > μ mg, giving tan θ < 1/(2μ). For μ = 0.45: tan θ_c = 1/0.9 = 1.11, so θ_c = 48°. **Check:** with a very rough floor (μ→∞) the ladder never slips, and with a frictionless floor it always does — both limits match the formula.

Solution 3 — Beam with two loads. Return →

(a) Take torques about the left support. The beam's weight 40g acts at 3.0 {m}; loads 60g at 1.5 {m} and 30g at 4.5 {m}. Then

$$6R_R = g(40\cdot 3.0 + 60\cdot 1.5 + 30\cdot 4.5) = g(120 + 90 + 135) = 345g, \tag{2}$$

so R_R = 57.5g = 564 {N}, and R_L = 130g - 57.5g = 72.5g = 711 {N}.

(b) A single 90 {kg} mass producing the same reactions must have the same total, which it does (90 = 40+60+30 is false — the total load is 130 {kg}, so no single 90 {kg} mass can reproduce the reactions; this is the point of the question). With the total load fixed at 130 {kg}, it must be placed where the torque sum is unchanged, at x with 130x = 345 x = 2.65 {m}.

(c) A non-uniform beam whose weight acts at a different point changes the torque balance but not the total vertical force; the reactions redistribute. **Check:** R_L + R_R = 130g exactly, as vertical balance demands.

Solution 4 — Three forces, one triangle. Return →

(a) F_1 + F_2 + F_3 = 0 is exactly the statement that they close a triangle when placed head to tail, by the definition of vector addition.

(b) If their lines of action were not concurrent, take torques about one intersection point of two of them: two torques vanish and the third cannot, contradicting equilibrium. Hence all three lines meet at a point.

(c) The triangle has interior angles 50° (at the 40° string), 25° and 105°. By the sine rule,

$$T_1 = (W\sin 65^\circ)/(\sin 105^\circ) = 0.939W,\ T_2 = (W\sin 40^\circ)/(\sin 105^\circ) = 0.665W. \tag{3}$$

Check: the string making the smaller angle with the horizontal carries the larger tension, which is the physical meaning of “nearly vertical is safe”.

Solution 5 — Sliding versus toppling. Return →

(a) Sliding starts when F = μ mg. Toppling starts when the weight line passes through the lower edge: about that edge, Fy = mg(w/2). Equal force at the threshold gives y_c = w/(2μ).

(b) For y > y_c the toppling force mgw/(2y) is smaller than the sliding force μ mg, so toppling wins.

(c) y_c = 0.35/(2×0.5) = 0.35 {m}; pushing at the top (y = 1.8 {m}) is far above this, so the shelf topples. **Check:** the criterion depends only on the ratio w/(2μ), so a wide enough shelf slides instead — the reason tall furniture is anchored.

Solution 6 — Hanging by a thread. Return →

(a) Vertical balance for the load with two symmetric tensions: 2Tsin θ = mg T = (mg)/(2sin θ).

(b) T 1200 {N} gives sin θ 245.3/(2400) = 0.102, so θ 5.9°.

(c) As θ→0, T→∞: a horizontal rope must supply an arbitrarily large tension to produce a finite vertical component. **Check:** at θ = 30°, T = mg = 245 {N}, so the rope is comfortable; the danger lies entirely in the small-angle regime, which is why clothes lines sag.

Solution 7 — Contact forces between stacked blocks. Return →

(a) The top block is driven by friction from the middle one: ma μ mg a μ g. So the whole stack's maximum common acceleration is μ g.

(b) The interface that fails first is the one with the smallest available friction relative to the mass it must accelerate. The upper interface must accelerate the top block only (μ mg available for mass m, so a μ g). The lower interface sees the middle block **plus** the top block, but the normal force there is 2mg if the blocks stack vertically and the friction available is μ(2mg), giving a μ g as well in the idealised model. With equal coefficients the two interfaces reach their limit together; if the blocks are stacked side by side against a vertical wall the normal forces differ and the lower pair slips first.

(c) Just before slip, a = μ g and F = 3mμ g (accelerating all three with the floor friction neglected or included as stated). **Check:** the answer scales linearly with g and μ, and F∝ m, both dimensionally required.

Solution 8 — Centre of mass of a composite shape. Return →

(a) Square area $4a^2$ centred at 0; hole area $\pi a^2/4$ at x = a. Treat the hole as negative mass:

$$x_{(cm)} = (4a^2 \cdot 0 - (\pi a^2/4)a)/(4a^2 - \pi a^2/4) = -(\pi a)/(4(4 - \pi/4)) \cdot (a)/(a) = -(\pi a)/(16-\pi) \approx -0.244a. \quad (4)$$

The lamina's centre of mass has moved **away** from the hole, as it must.

(b) For an added half-disc of radius a on the right edge, its centroid is 4a/(3π) from the flat edge, so at x = a + 4a/(3π) = 1.424a with area $\pi a^2/2$. Then

$$x_{(cm)} = (4a^2 \cdot 0 + (\pi a^2/2)(1.424a))/(4a^2 + \pi a^2/2) = (0.712\pi a)/(4+\pi/2) \approx 0.400a. \quad (5)$$

(c) As the hole radius →0, both results tend to 0; as the hole grows to the full square, the first formula diverges, signalling that the remaining lamina is not a valid body to balance. **Check:** signs are opposite for removal and addition, as they must be.

Solution 9 — Small oscillations from a potential. Return →

(a) U'(x) = U_0(-(a)/(x²) + (a²)/(x³)) = (U_0 a)/(x³)(a - x). Zero at x = a.

(b) U''(x) = U_0((2a)/(x³) - (3a²)/(x^4)), so U''(a) = U_0((2)/(a²) - (3)/(a²)) = -(U_0)/(a²) < 0. The second derivative is negative, so the stationary point is a **maximum** and therefore unstable, not stable. A particle displaced from x=a runs away; the “oscillation” question is a trap, and the correct conclusion is that no stable small oscillation exists about x=a.

(c) U(x) = U_0((a)/(x) - (a²)/(2x²)) tends to 0 as x→∞ and to -∞ as x→0^+, with a maximum U_0/2 at x = a. A particle with E > U_0/2 therefore comes in from infinity, passes the barrier and falls into x→0; with E it turns around at the outer turning point. **Check:** U'(a)=0 and U''(a)<0 are consistent, and the shape - rising from -∞, peaking, decaying to 0 — matches the algebra.

Solution 10 — A tripod. Return →

(a) The three legs are equivalent by a 120° rotation, so their vertical components must be equal: 3Ncos α = W Ncos α = W/3 per leg, where N is the force along each leg.

(b) Each leg's force has a horizontal inward component Nsin α = (Wtan α)/(3) directed toward the axis; the three horizontal components cancel by symmetry, but the head still feels an inward squeeze.

(c) If the legs pointed inward the ground would have to **pull**, which it cannot, so the legs must splay outward and the ground pushes inward on the feet. The horizontal force per foot is (Wtan α)/(3), which friction must balance: μ (tan α)/(3). **Check:** with α = 20° this needs μ 0.12, which is why tripods have rubber feet and why a very splayed tripod creeps on smooth floors.